

Silver Maple

Acer saccharinum

Species background: Native to eastern North America, especially along rivers and floodplains, where its fast growth and flood tolerance make it well-suited to unstable soil. Its leaves are the most deeply lobed of the common street maples, cut almost to the midrib, and the pale undersides create a shimmering, two-toned effect whenever wind moves through the canopy.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Silver Maple: Notice silver maple's flashing silver leaf undersides on a windy day.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, silver and green crayons.

Teacher background

Wait for a windy day and watch my leaves flip -- the undersides are silvery-white!

Procedure

- 1 Engage: Ask what color students think the bottom of a leaf might be.
- 2 Explore: If it's windy, watch the leaves flip and flash silver; otherwise turn one over by hand.
- 3 Explain: Explain that this tree's leaf undersides are much paler than its green tops.
- 4 Art: Turn a leaf over and color one side green, one side silver.
- 5 Evaluate: Student can describe the leaf as green on top and silver underneath.

Discussion questions

What other things look different from two sides?

Assessment

Correctly describes the two-toned leaf.

Extension

Look for silver flashes in the treetops on a windy walk.

Silver Maple: Compare silver maple's growth speed and leaf shape to another common maple.

Standards connection: NGSS 3-LS4-3; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

It's one of the fastest-growing large trees in North America. Its leaves are deeply cut, almost to the midrib -- more deeply lobed than most other city maples.

Procedure

- 1 Engage: Ask what might be good and bad about a tree that grows very fast.
- 2 Explore: Compare a silver maple leaf's deep cuts to another maple leaf if one is nearby.
- 3 Explain: Discuss the tradeoff between fast growth and weaker wood.
- 4 Art: Paint or draw one leaf with two colors -- green on one side, silvery-white on the other, just like the real thing in the wind.
- 5 Evaluate: Student writes one sentence naming a tradeoff of fast growth.

Discussion questions

Would you rather plant a fast-growing tree or a slow, sturdy one on your street? Why?

Assessment

Correct sentence naming the fast-growth/weak-wood tradeoff.

Extension

Measure and compare leaf lobe depth between silver maple and another maple species.

GRADE BAND 6-8

Silver Maple: Explain the tradeoff between growth rate and wood density using silver maple and oak as a comparison.

Standards connection: NGSS MS-LS1-5; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, two-tone art materials.

Teacher background

Silver maple grows fast with wide, weak wood; oaks grow slowly with dense, strong wood -- a useful comparison to make on the same walk. Its winged seeds are the largest of any North American maple.

Procedure

- 1 Engage: Why might a tree 'choose' fast growth over strong wood, in evolutionary terms?
- 2 Explore: Research the relationship between growth rate and wood density across tree species.
- 3 Explain: Compare silver maple's strategy to a slow-growing oak's.
- 4 Art: Create a two-tone flip-art piece: green on one side, silver on the other, mounted so both sides are visible -- write a caption explaining why the underside is pale.
- 5 Evaluate: Write a paragraph comparing the two growth strategies and their real-world tradeoffs for a city street tree.

Discussion questions

Why might a city choose to plant fewer fast-growing trees and more slow, sturdy ones -- or vice versa?

Assessment

Paragraph correctly compares both growth strategies with specific tradeoffs.

Extension

Research a real municipal storm-damage report and see which tree species are most commonly involved.

Silver Maple: Evaluate growth-strategy tradeoffs using a real municipal tree-risk assessment framework.

Standards connection: NGSS HS-LS2-6; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, camera or sketching materials for a diptych.

Teacher background

Silver maple prioritizes fast growth with low-density wood, making it more prone to storm breakage than slower-growing oaks. Research a real municipal tree-risk assessment framework and evaluate what risk factors an arborist would flag for this species versus a mature oak on the same street.

Procedure

- 1 Engage: What factors would a professional arborist weigh when assessing storm risk for a specific tree?
- 2 Explore: Research a real urban forestry risk-assessment framework and its criteria.
- 3 Explain: Apply the framework's criteria to compare silver maple and a mature oak.
- 4 Art: Photograph the leaf from both sides under the same lighting, then create a composite diptych analyzing how light reflectance differs, citing the physical mechanism (waxy cuticle vs. trichome density) responsible.
- 5 Evaluate: Submit the risk-assessment comparison and the diptych with its written analysis.

Discussion questions

Should storm-risk concerns outweigh the shade and cooling benefits of fast-growing trees in urban planning decisions?

Assessment

Rubric covering correct application of the risk framework and diptych analysis quality.

Extension

Compare silver maple's risk profile to London planetree's storm resilience.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.